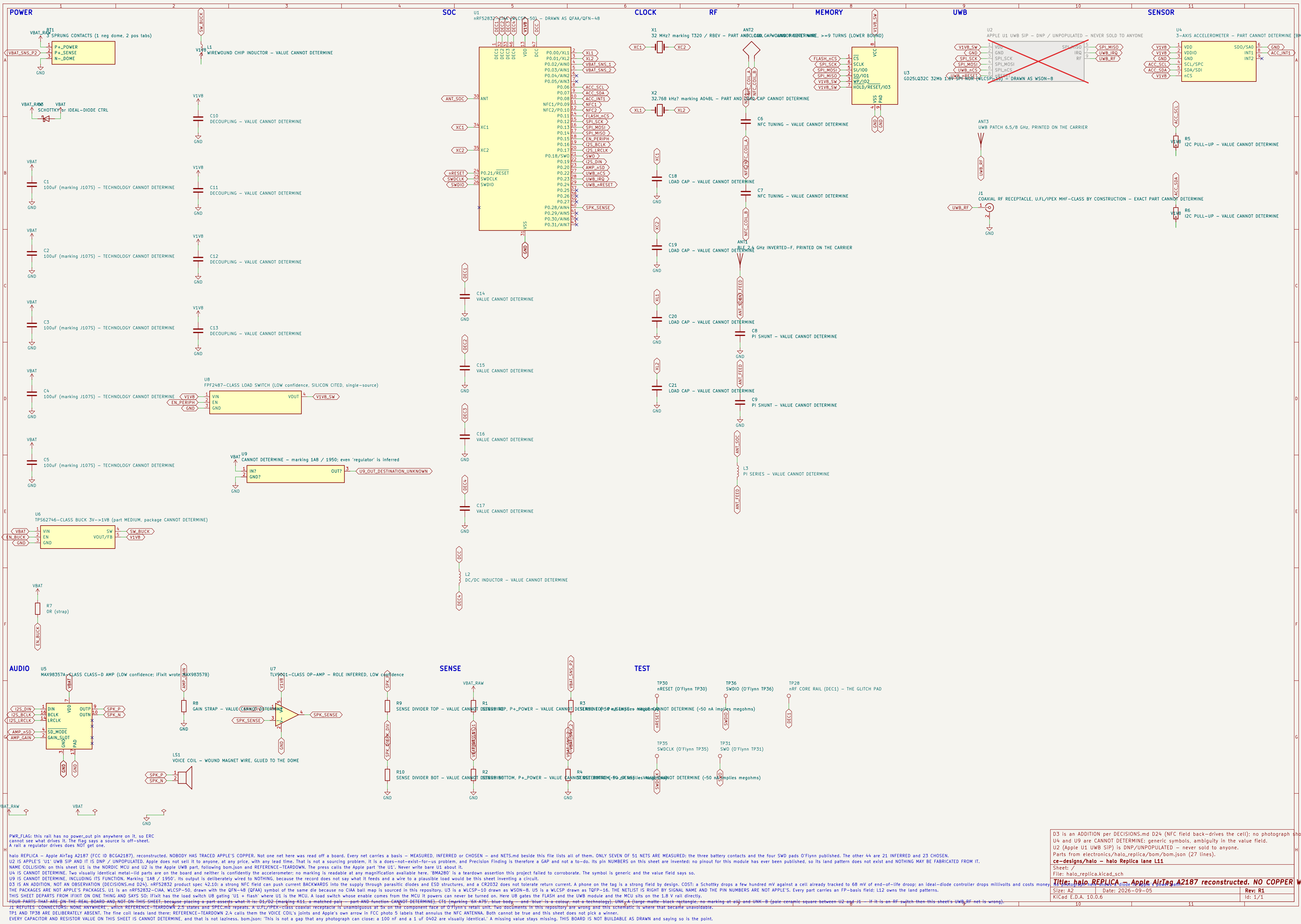




PWR_FLAG: this rail has no power_out pin anywhere on it, so ERC cannot see what drives it. The flag says a source is off-sheet. A rail a regulator drives does NOT get one.

halo REPLICA – Apple AirTag A2187 (FCC ID BGA2187), reconstructed. NOBODY HAS TRACED APPLE’S COPPER. Not one net here was read off a board. Every net carries a basis – MEASURED, INFERRED OR CHOSEN – and NETS.md beside this file lists all of them. ONLY SEVEN OF 51 NETS ARE MEASURED: the three battery contacts and the four SWD pads O’Flynn published. The other 44 are 21 INFERRED and 23 CHOSEN.

U2 IS APPLE’S ‘U1’ UWB SIP AND IT IS DNP / UNPOPULATED. Apple does not sell it to anyone, at any price, with any lead time. That is not a sourcing problem, it is a does-not-exist-for-us problem, and Precision Finding is therefore a GAP and not a to-do. Its pin NUMBERS on this sheet are invented: no pinout for this module has ever been published, so its land pattern does not exist and NOTHING MAY BE FABRICATED FROM IT.

NAME COLLISION: on this sheet U1 is the NORDIC MCU and U2 is the Apple UWB part, following bom,json and REFERENCE-TEARDOWN. The press calls the Apple part ‘the U1’. Never write bare U1 about it.

U4 IS CANNOT DETERMINE. Two visually identical metal-lead parts are on the board and neither is confidently the accelerometer: no marking is readable at any magnification available here. ‘BMA280’ is a teardown assertion this project failed to corroborate. The symbol is generic and the value field says so.

U9 IS CANNOT DETERMINE, INCLUDING ITS FUNCTION. Marking ‘1A8 / 1950’. Its output is deliberately wired to NOTHING, because the record does not say what it feeds and a wire to a plausible load would be this sheet inventing a circuit.

D3 IS AN ADDITION, NOT AN OBSERVATION (DECISIONS.md D24). nRF52832 product spec 42.10: a strong NFC field can push current BACKWARDS into the supply through parasitic diodes and ESD structures, and a CR2032 does not tolerate return current. A phone on the tag is a strong field by design. COST: a Schottky drops a few hundred mV against a cell already tracked to 68 mV of end-of-life droop; an ideal-diode controller drops millivolts and costs money.

THE PACKAGES ARE NOT APPLE’S PACKAGES. U1 is an nRF52832–CIAA, WLCS–50, drawn with the QFN–48 (QFAA) symbol of the same die because no CIAA ball map is sourced in this repository. U3 is a WLCS–10 drawn as WSON–8. U5 is a WLCS–16. THE NETLIST IS RIGHT BY SIGNAL NAME AND THE PIN NUMBERS ARE NOT APPLE’S. Every part carries an FP–basis field; L12 owns the land patterns. THIS SHEET DEPARTS FROM IFIXIT ON ONE THING AND SAYS SO: ifixit has the load switch U8 gating ‘U1 + flash’ where U1 is the MCU. A load switch whose enable comes from the MCU it powers can never be turned on. Here U8 gates the FLASH and the UWB module and the MCU sits on the 1.8 V rail directly.

FOUR PARTS THAT ARE ON THE REAL BOARD AND NOT ON THIS SHEET, because placing a part asserts what it is: D14–D22 (marking K11; a matched pair – part AND function CANNOT DETERMINE); C14 (marking 6X-A75; blue body – and blue is a colour, not a technology); UNK-A (large matte black rectangle, no marking at all) and UNK-B (pale ceramic square between U2 and J1 – if it is an RF switch then this sheet’s UWB_RF net is wrong).

J1 REPUTES ‘CONNECTORS: NONE ANYWHERE’, with REFERENCE-TEARDOWN 2.25 states and SPEC.md repeats: ‘A U.FL/IPEX–class coaxial receptacle’ is ‘unambiguous’ at 5x on the component face of O’Flynn’s test: ‘Two documents in this repository are wrong and this schematic is where that became unavoidable.’

TP1 and TP38 ARE DELIBERATELY ABSENT. The fine coil leads land there: REFERENCE-TEARDOWN 2.4 calls them the VOICE COIL’s joints and Apple’s own arrow in FCC photo 5 labels that annulus the NFC ANTENNA. Both cannot be true and this sheet does not pick a winner.

EVERY CAPACITOR AND RESISTOR VALUE ON THIS SHEET IS CANNOT DETERMINE, and that is not laziness. bom,json: ‘This is not a gap that any photograph can close: a 100 nF and a 1 uF 0402 are visually identical.’ A missing value stays missing. THIS BOARD IS NOT BUILDABLE AS DRAWN and saying so is the point.

D3 is an ADDITION per DECISIONS.md D24 (NFC field back–drives the cell); no photograph shows one.

U4 and U9 are CANNOT DETERMINE: generic symbols, ambiguity in the value field.

U2 (Apple U1 UWB SIP) is DNP/UNPOPULATED – never sold to anyone.

Parts from electronics/halo_replica/bom/bom.json (27 lines).

ce-designs/halo – halo Replica lane L11

Sheet: /

File: halo_replica.kicad_sch

Title: halo_REPLICA – Apple AirTag A2187 reconstructed. NO COPPER WAS TRACED

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